MILITARY SPECIFICATION

CARTRIDGE, 40MM, STAR, PARACHUTE

XM661 AND XM662

LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the metal parts and loading, assembling and packing for two types of cartridges designated as Cartridges: 40MM, Star, Parachute, XM661 and XM662.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-223	- Black Powder
MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-I-45607	- Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

FSC: 1310

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MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-331	-	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1234	-	Pyrotechnics Sampling, Inspection and Testing
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

9255145	-	40MM, Star, Parachute, Cartridge Assembly

PUBLICATIONS
ARMY

ET-9255145	-	Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49	-	Transportation, Parts 0-190

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify, "49 CFR 0-190 (latest revision)".)

3. REQUIREMENTS

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3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Cartridge.-The cartridge shall comply with all requirements specified on Drawing (Dwg.) 9255145, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Transportation vibration.-The cartridge shall comply with the following requirements when tested as specified in 4.4.5.

3.3.1 The cartridge shall be safe to transport following this test (See 6.6).

3.3.2 There shall be no evidence of external damage to the cartridge that will affect the intended function (See 6.7).

3.4 Functioning.-The cartridge shall comply with the following requirements.

3.4.1 Altitude.-The projectile shall function at an average altitude of 600 ± 100. feet.

3.4.2 Burning of illuminate assembly.-The illuminant assembly shall burn in the air for 30 seconds min. while suspended from the parachute.

3.4.3 Parachute.-The parachute shall not separate from the illuminant assembly or delay opening from time of ejection for more than 5 seconds.

3.4.4 Firing.-No projectile shall stick in the gun bore.

3.4.5 Hot temperature (160 degrees F. plus or minus 5 degrees (F.).-The cartridge shall comply with the requirements of 3.4.1, 3.4.2, 3.4.3 and 3.4.4.

3.4.6 Ambient temperature (70 degrees F. plus or minus 5 degrees F.).-The cartridge shall comply with the requirements of 3.4.1, 3.4.2, 3.4.3 and 3.4.4.

3.4.7 Cold temperature (-65 degrees F. plus or minus 5 degrees F.).-The cartridge shall comply with the requirements of 3.4.1, 3.4.2, 3.4.3 and 3.4.4.

3.5 Workmanship.-All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner.

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They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required marking and stamping shall be neat and sharply defined.

3.6 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (see 6.5).

3.7 First article testing.--This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

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d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.

4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1). The first article sample shall consist of the following:

a. Twenty-five (25) illuminant assemblies (See Dwg. 9252412).
b. Twenty-five (25) cartridge assemblies (See Dwg. 9255145).
c. Four-hundred and forty (440) cartridges (See Dwg. 9255145) to a Government proving ground for functioning tests.
d. Twenty-five (25) delay loading assemblies (See Dwg. 9255409).

All parts and completed assemblies submitted shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspection to be performed.-All assemblies shall be subjected by the Government to any or all the examinations or tests specified in this specification and any or all requirements of the applicable drawings.

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4.2.3 Rejection.-If any assembly fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.2.4 Remaining first article inspection samples.-Unless otherwise specified by the Contracting Officer, loaded cartridges shall not be submitted for proving ground tests until the subassemblies and assemblies submitted in accordance with 4.2 have been accepted.

4.2.4.1 Prior to submission.-Prior to submission of the four hunder and forty (440) cartridges to a Government proving ground for functioning tests, the 440 cartridges shall be inspected at the contractor's plant for the following tests:

4.2.4.1.1 Transportation vibration (See 3.3).-The 440 cartridges packed in their regular shipping containers shall be subjected to this test.

4.2.4.1.2 Air pressure (See Dwg. 9255145).-After completion of test specified in 4.2.4.1.1, the 440 cartridges shall be removed from their regular shipping containers and subjected to this test.

4.2.5 Rejection.-If any assembly fails to comply with any of the applicable requirement, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure to an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.2.6 Proving ground tests.-The cartridges shall be subjected by the Government to the following tests:

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a. Functioning at hot temperature (see 3.4.5).-Eighty (80) cartridges shall be subjected to this test (see 4.4.6).
b. Functioning at ambient temperature (See 3.4.6).-Two-hundred and eighty (280) cartridges shall be subjected to this test (See 4.4.6).
c. Functioning at cold temperature (see 3.4.7).-Eighty (80) cartridges shall be subjected to this test (see 4.4.6).

4.2.6.1 Rejection.-The first article sample shall be rejected if:

a. Any critical defect occurs in any phase (see Table III).
b. Average altitude of the total sample fails the applicable requirement (see Table III).
c. A total of eight (8) or more in the hot phase fail any of the following:
(1) Parachute failure (see 3.4.3).
(2) Item fails to eject (See Table III).
(3) No illumination (See Table III).
Burning time should be recorded for informational purposes (average).
d. A total of twenty-two (22) or more in the ambient phase fail any of the following:
(1) Burning time incorrect (see 3.4.2). (Average)
(2) Parachute failure (see 3.4.3).
(3) Item fails to function (see Table III).
e. A total of eight (8) or more in the cold phase fail any of the following:
(1) Parachute failure (see 3.4.3).
(2) Item fails to function (see Table III).
(3) No illumination (See Table III).
Burning time should be recorded for informational purposes (Average).
f. Two (2) or more of the total sample are below an altitude of 350 ft (see Table III).

In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity. Until the first article quantity is accepted, the contractor is not authorized by the Government to proceed with regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and in-

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spected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall contain:

a. Projectile metal parts from one interfix lot number from one manufacturer.
b. Delay composition from not more than one lot number.
c. Igniter composition from not more than one lot number.
d. Liner composition from not more than one lot number.
e. Illuminant composition produced by one manufacturer under one continuous set of operating conditions and which consist of one or more batches that have been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous.
f. Cartridge case loaded assemblies from one interfix lot number from one manufacturer.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Illuminant assembly (See Dwg. 9252412 covering a detail of Dwg. 9252411).-

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined		
Major	AQL 0.65 Percent		
101.	Anchor plug above to below flush	Gage	01001
102.	Diameter, maximum	Gage	01002
103.	Anchor plug insecure	Manual-Visual	01003
104.	Assembly damaged to extent that function may be impaired	Visual	01004
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.5)	Visual	01005

4.3.2.2 Projectile assembly, prior to assembling illuminant assembly (See Dwg. 9252411 covering a detail of Dwg. 9255145).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Coupling of illuminant assembly not secure to chain	Visual	02001
Minor:	None defined		

4.3.2.3 Projectile assembly, prior to assembling ogive (See Dwg. 9252411 covering a detail of Dwg. 9255145).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Parachute missing	Visual	03001
102.	O-ring missing	Visual	03002
Minor:	None defined		

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4.3.2.4 Projectile assembly (See Dwg. 9252411 covering a detail of Dwg. 9255145).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Gap present between delay assembly and body	Visual	04001
102.	Incorrect color	Visual	04002
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.5)	Visual	04003

4.3.2.5 Cartridge (See Dwg. 9255145).-

Categories	Defects	Method of Inspection	Code No.
Critical			
1.	Chamber gage failure	Gage	05001
2.	Incorrect ogive	Visual	05002
3.	Marking incorrect	Visual	05003
Major	AQL 0.40 Percent		
101.	Marking unidentifiable	Visual	05004
Minor	AQL 1.00 Percent		
201.	Total length	Gage	05005
202.	Evidence of poor workmanship (See 3.5)	Visual	05006

4.3.2.6 Unsealed metal box (See Dwg. 9209204).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Incorrect number of cartridges	Visual	06001
102.	Desiccant bags missing	Visual	06002
103.	Rubber gasket damaged or improperly assembled	Visual	06003
Minor:	AQL 1.00 Percent		
201.	Fillers missing	Visual	06004
202.	Tube missing	Visual	06005

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4.3.2.7 Sealed metal box (See Dwg. 9209204).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 1.00 Percent		
201.	Marking misleading or unidentifiable	Visual	07001
202.	Contents loose	Visual	07002

4.3.2.8 Sealed wooden packing box (See Dwg. 9209205).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 1.50 Percent		
201.	Marking misleading or unidentifiable	Visual	08001
202.	Contents loose	Manual	08002
203.	ICC nomenclature missing	Visual	08003

4.3.3 Testing.-

4.3.3.1 Moisture content (See TABLE I), Major defect.-

TABLE I

Material	Code No.
Delay Composition (See Dwg. 9252409)	09001
Igniter Composition (See Dwg. 9252409)	09002
Illuminant composition (See Dwg. 9252412)	09003
Ignition composition (See Dwg. 9252412)	09004
Black powder (See Dwg. 9252412)	09005

The contractor shall provide adequate controls to insure compliance with the requirements and shall test for verification at least one sample of each material from each eight hour' production of cartridges. Composite samples shall not be used. If the moisture content of a sample exceeds the requirement,

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and loading has not begun, that quantity of material represented by the sample shall be rejected. If assemblies have been loaded with material containing excessive moisture, the remaining material that is represented by the sample together with all cartridges loaded with the non-conforming material shall be rejected. The tests shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4.

4.3.2.2 Static burning time of delay loading assembly (See Dwg. 9252409), Major Defect, Code No. 10001.-Thirty-two (32) delay loading assemblies shall be selected from each lot and subjected to this test. If two (2) or more assemblies fail to comply with the requirement specified on the applicable drawing, the lot shall be rejected. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4.

4.3.2.3 Average static candle power of illuminant loading assembly (See Dwg. 9252412), Major defect, Code No. 11001.-Thirty-two (32) illuminant loading assemblies shall be selected from each lot and subjected to this test. If the average candle power fails to comply with the requirement specified on the applicable drawing, the lot shall be rejected. The test shall be performed as specified in 4.4.3 using equipment in accordance with 4.3.4.

4.3.3.4 Security of projectile assembly (See Dwg. 9255145) Major Defect, Code No. 12001).-The sampling plan for this test shall be in accordance with MIL-STD-105, Code Letter H, single plan, with an AQL of 2.50 percent. The test shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4.

4.3.3.5 Air pressure (See Dwg. 9255145), Major Defect, Code No. (13001).-This test shall be conducted 100 percent. Any cartridge which fails to comply with the requirement specified on the applicable drawing shall be classified defective and removed from the lot. The test shall be performed as specified in 4.4.5 using equipment in accordance with 4.3.4.

4.3.3.6 Transportation vibration (See 3.3), Major Defect, Code No. (14001).-The cartridges shall be observed and examined visually without disassembly for any evidence of failure to comply with the requirements as classified in Table II when tested as specified in 4.4.6.

TABLE II

Defect	Classification	Code No.
Cartridge not safe to transport following test (see 3.3.1)	Critical	14002

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Cartridge damaged after test (see 3.3.2)	Major	14003

4.3.3.6.1 After completion of test specified in 4.3.3.5 and beginning with the first lot produced and continuing until three (3) acceptable lots (lots not accepted by disposition) have complied with the acceptance criteria specified in Table II, two hundred and sixty-four (264) cartridges shall be selected from each lot, packed in their regular shipping containers and subjected to this test. The lot shall be rejected if any defect as classified in Table II occurs.

4.3.3.7 Air pressure (See Dwg. 9255145), Major Defect, Code No. 15001).-After completion of test specified in 4.3.3.6.1, the two hundred sixty four (264) samples from each lot shall be subjected to this test. If nine (9) or more cartridges fail to comply with the requirements specified on the applicable drawing, the lot shall be rejected. The test shall be performed as specified in 4.4.5 using equipment in accordance with 4.3.4. Cartridges which fail this test shall be discarded and replaced with units which have been subjected to test specified in 4.3.3.6 prior to subsequent testing.

4.3.3.8 Functioning (See 3.4), Major Defect, Code No. (16001).- The cartridges shall be observed for any evidence of failure to comply with the requirements as classified in Table III when tested as specified in 4.4.7.

TABLE III

Defect	Classification	Code No.
Any projectile assembly sticks in the gun bore (see 3.4.4)	Critical	16002
Burning time incorrect (see 3.4.2)	Major	16003
The first emission of light shall be at an average altitude of 600 ± 100 ft.	Major	16004
Parachute failure (see 3.4.3)	Major	16005
Altitude less than 350 ft.	Major	16006
Item fails to function	Major	16007

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4.3.3.8.1 Hot temperature.-After completion of test specified in 4.3.3.7 and beginning with the first lot produced and continuing until three (3) acceptable lots (lots not accepted by disposition) have complied with the acceptance criteria specified in Table III, forty-four (44) of the samples from each lot shall be subjected to this test.

4.3.3.8.2 Ambient temperature.-After completion of test specified in 4.3.3.7 and beginning with the first lot produced and continuing until three (3) acceptable lots (lots not accepted by disposition) have complied with the acceptance criteria specified in Table III, one hundred and seventy-six (176) of the samples from each lot shall be subjected to this test.

4.3.3.8.3 Cold temperature.-After completion of test specified in 4.3.3.7 and beginning with the first lot produced and continuing until three (3) acceptable lots (lots not accepted by disposition) have complied with the acceptance criteria specified in Table III, forty-four (44) of the samples from each lot shall be subjected to this test.

4.3.3.8.4 Rejection.-The lot shall be rejected if:

a. Any critical defects occurs in any phase (see Table III).
b. Average altitude of the total sample fails the applicable requirement (see Table III).
c. A total of six (6) or more in the hot phase fail any of the following:
(1) Parachute failure (see 3.4.3)
(2) Item fails to function (see Table III).
Burning time should be recorded for informational purposes. (Average).
d. A total of fifteen (15) or more in the ambient phase fail any of the following:
(1) Burning time incorrect (see 3.4.2). (Average)
(2) Parachute failure (see 3.4.3).
(3) Item fails to function (see Table III).
e. A total of six (6) or more in the cold phase fail any of the following:
(1) Parachute failure (see 3.4.3).
(2) Item fails to function (see Table III).
Burning time should be recorded for informational purposes. (Average).
f. Two (2) or more of the total samples have an altitude less than 350 ft (see Table III).

4.3.3.9 Transportation vibration (see para. 3.3), Regular Production, Major Defect, Code No. (17001).-The cartridge shall be observed and examined visually without disassembly for any evidence of failure to comply with the requirements as classified in Table II when tested as specified in 4.4.6.

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4.3.3.9.1 After three (3) acceptable lots have complied with 4.3.3.6.1, and 4.3.3.8 one hundred and eighty-nine (189) of the samples subjected to test specified in 4.3.3.6 shall be selected from each lot, packed in their regular shipping containers and subjected to this test. The lot shall be rejected if any defect as classified in Table II occurs.

4.3.3.10 Air pressure (see dwg. 9255145), Major Defect, Regular Production, Code No. (18001).-After completion of test specified in 4.3.3.9.1, the one hundred and eighty-nine (189) samples shall be removed from their shipping containers and subjected to this test. If eight (8) or more samples fail to comply with the requirement specified on the applicable drawing, the lot shall be rejected. The test shall be performed as specified in 4.4.5 using equipment in accordance with 4.3.4. Cartridges which fail this test shall be discarded and replaced with units which have been subjected to test specified in 4.3.3.9 prior to subsequent testing.

4.3.3.11 Functioning (see 3.4), Regular Production, Major Defect, Code No. (19001).-The cartridges shall be observed for any evidence of failure to comply with the requirements as classified in Table III when tested as specified in 4.4.6.

4.3.3.11.1 Hot temperature.-After three (3) acceptable lots have complied with 4.3.3.8, twenty-six (26) of the samples subjected to test specified in 4.3.3.11 shall be subjected to this test.

4.3.3.11.2 Ambient temperature.-After three (3) acceptable lots have complied with 4.3.3.8, eighty (80) of the samples subjected to test specified in 4.3.3.10 shall be subjected to this test.

4.3.3.11.3 Cold temperature.-After three (3) acceptable lots have complied with 4.3.3.8, twenty-six (26) of the samples subjected to test specified in 4.3.3.10 shall be subjected to this test.

4.3.3.11.4 Rejection.-The lot shall be rejected if:

a. Any critical defect occurs in any phase (see Table III).
b. Average altitude of the total sample fails the applicable requirement (see Table III).
c. A total of four (4) or more in the hot phase fail any of the following:
(1) Parachute failure (see 3.4.3).
(2) Item fails to function (see Table III).
Burning time should be recorded for informational purposes. (Average).

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d. A total of eight (8) or more in the ambient phase fail any of the following:

(1) Burning time incorrect (see 3.4.2). (Average)
(2) Parachute failure (see 3.4.3).
(3) Item fails to function (see Table III).

e. A total of four (4) or more in the cold phase fail any of the following:

(1) Parachute failure (see 3.4.3).
(2) Item fails to function (see Table III).

Burning time should be recorded for informational purposes. (Average).

f. Two (2) or more of the total samples have an altitude less than 350 ft (see Table III).

If any lots fails, the consecutive lots thereafter shall revert to the criteria of 4.3.3.8.

4.3.3.12 Suitability test plan (Informantion test only), Code No. (20001).-Beginning with the first lot produced by each producer and continuing until that producer has submitted three (3) acceptable lots (lots not accepted by disposition) that are accepted in accordance with 4.3.3.8.1, 4.3.3.8.2 and 4.3.3.8.3, and extra sample of eighty-eight (88) rounds shall be selected from each lot and preconditioned and tested as follows:

a. The eighty-eight (88) rounds shall be packaged in their regular shipping containers and subjected to transportation vibration environmental in accordance with MIL-STD-331, Test No. 104.

b. After transportation vibration, the rounds shall be unpacked and subjected to the JAN temperature and humidity cycle (14 days) specified in MIL-STD-331, Test No. 105.

c. After temperature and humidity cycle, the rounds shall be temperature conditioned and fixed as follows:

30 rounds +165 degrees F. ± 5 degrees F.
28 rounds ambient (between +40 and +110 degrees F.)
30 rounds -65 degrees F. ± 5 degrees F.

d. Fire the rounds in accordance with paragraph 4.4.7 and with TECOM Material Test Procedure, TECP 700-700 Volume 4. Test results, failure analysis reports, and other information as requested shall be forwarded to Picatinny Arsenal, ATTN: SMUPA-QA-A-S.

4.3.3.13 Air pressure test of packed, sealed metal box (see dwg. 9209204), Major defect, Code No. (28001).-One (1) packed, sealed metal box shall be selected from each twenty (20)

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boxes packed. If the metal box fails to comply with the requirement specified on the applicable drawing, the quantity represented by the sample shall be rejected. The test shall be performed as specified in 4.4.8 using equipment in accordance with 4.3.4.

4.3.4 Inspection equipment.-Equipment Tabulation (ET) number 9255145 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.4 Test methods and procedures.-

4.4.1 Moisture content.-

4.4.1.1 Delay composition.-The moisture content of the delay composition at the time and place of loading shall be determined in accordance with Method 102.1 of MIL-STD-1234.

4.4.1.2 Black powder.-The moisture content of the black powder at the time and place of loading shall be determined in accordance with 4.4.3 of MIL-P-223.

4.4.1.3 Ignition composition.-The moisture content of the ignition composition at the time and place of loading shall be determined in accordance with Method 101.4 of MIL-STD-1234 using methyl alcohol as an extraction solvent.

4.4.1.4 Illuminant composition.-The Karl Fischer method as stated in Standard MIL-STD-1234, Method 101.2 up to paragraph 5.3 shall be used. A fifty-gram sample plus or minus 0.1 gram shall be added to a 500 milliliter (ml) volumetric flash containing approximately 300 to 400 ml. methanol and 25 grams of dry sodium nitrate. The flask shall be stoppered and the contents

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swirled cautiously for several minutes until the material is thoroughly dispersed. The sample shall be allowed to remain in contact with the methanol for approximately two hours. Then the 500 ml. volumetric flask shall be filled up to the 500 ml. mark with methanol and swirled again. A blank without the sample shall be put through the same procedure. After the sample has settled, a 50 ml. aliquot of the clear supernatant liquid shall be withdrawn and immediately transferred to the standard titration vessel containing approximately 100 ml. of methanol which has just been titrated to the preliminary end point as described in Standard MIL-STD-1234, Method 101.2, paragraph 5.1. The final end point shall be reached in 3.5 minutes in the manner described in Standard MIL-STD-1234, Method 101.2, paragraph 5.4. A 50 ml. aliquot of the blank shall be titrated in the same manner. The water content shall be calculated as follows:

$$\% \text{ Water} = \frac{100F\,[(VR-S) - (V'R-S')]}{W}$$

where:

F = grams of water per ml. of standard water in methanol solution

V = ml. of Karl Fischer reagent added to the sample

V' = ml. of Karl Fischer reagent added to the blank

R = ml. of standard water in methanol solution per ml. of Karl Fischer reagent

S = ml. of standard water in methanol solution for titration of sample

S' = ml. of standard water in methanol solution with back titration of blank

W = weight of sample in grams

4.4.1.5 Igniter composition.-The moisture content of the igniter composition at the time and place of loading shall be determined in accordance with 4.6.4 of MIL-P-22264.

4.4.2 Static burning time of delay loading assembly.-This test shall be conducted in accordance with the procedure shown on the Equipment List.

4.4.3 Average static candle power of illuminant assembly.-This test shall be conducted in accordance with the procedure shown on the Equipment List.

4.4.4 Security of projectile assembly.-The cartridge shall be placed in the fixture shown on the Equipment List and the axial force specified on dwg. 9255145 shall be applied. Assembly shall be pulled until total separation occurs and data shall be recorded. This test is a destructive test. Parts so tested shall not be returned to the lot.

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4.4.5 Air pressure.-The cartridge shall be placed in the fixture shown on the Equipment List and a measured quantity of air shall be applied to produce the required air pressure. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.6 Transportation vibration.-The cartridges shall be packaged and packed in accordance with dwgs. 9209204 and 9209205 and tested in accordance with Test No. 104 of MIL-STD-331, except that each box shall be vibrated at the specified amplitudes for four (4) hours in each of three (3) different positions (i.e., box positioned so that the cartridges are vertical with case end down, box positioned so that the cartridges are horizontal, and box positioned so that the cartridges are vertical with case end up. After the test, the box packing and the cartridges shall be examined to determine compliance with the requirements. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.7 Functioning.-The following tests shall be performed at a Government-owned proving ground:

4.4.7.1 Hot temperature (160 degrees F. plus or minus 5 degrees F.).-The cartridges shall be packed in their regular shipping containers and conditioned for a minimum of 16 hours at a temperature of 160 degrees F. plus or minus 5 degrees F. The cartridge shall be taken from the conditioning chamber, removed from the packing container, and fired from an approved launcher placed on an approved mount at 90 degrees quadrant within approximately 2 minutes. Observation shall be made for proper functioning and the requirements of 3.4.5.

4.4.7.2 Ambient temperature (70 degrees F. plus or minus 5 degrees F.).-The cartridges shall be packed in their regular shipping containers and conditioned for a minimum of 16 hours at a temperature of 70 degrees F. plus or minus 5 degrees F. The cartridge shall be taken from the conditioning chamber, removed from the packing container, and fired from an approved launcher placed on an approved mount at 90 degrees quadrant within approximately 2 minutes. Observation shall be made for proper functioning and the requirements of 3.4.6.

4.4.7.3 Cold temperature (-65 degrees F. plus or minus 5 degrees F.).-The cartridges shall be packed in their regular shipping containers and conditioned for a minimum of 16 hours

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at a temperature of -65 degrees F. plus or minus 5 degrees F. The cartridge shall be taken from the conditioning chamber, removed from the packing container, and fired from an approved launcher placed on an approved mount at 90 degrees quadrant within approximately 2 minutes. Observation shall be made for proper functioning and the requirements of 3.4.7.

4.4.8 Air pressure of packed, sealed metal box.-The metal box shall be placed in the fixture shown on the Equipment List and a measured quantity of air shall be applied to produce the required air pressure. Observation shall be made for leakage. This test is a non-destructive test. Boxes so tested may be returned to the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level A.-The cartridges shall be preserved and packaged in accordance with dwg. 9209204.

5.2 Packing.-

5.2.1 Level A.-The cartridges, preserved and packaged as specified in 5.1.1, shall be packed in accordance with dwg. 9209204.

5.3 Marking.-Marking shall be in accordance with dwgs. 9209204, and 9209205, and Code of Federal Regulations, Title 49, Parts 0-190.

5.4 Shipping.-When shipments from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of First Article Samples.

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6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs submitted by the contractor (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Equipment lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010) the number of drawings for Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army designs, the contractor shall comply with and

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use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor-prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes, and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal ATTN: SMUPA-QA-A-S. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

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6.4 Submission of test data.-In addition to the normal distribution of records, when the cartridges are procured by the U.S. Army Munitions Command, one (1) copy of all ballistic data shall be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-S.

6.5 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-S.

6.6 The cartridges will be considered safe to transport providing no evidence exists of loose powder or composition in the box.

6.7 The cartridges will be considered free of damage that will affect the intended function provided the top seal has not been broken by movement or displacement of the top, or the case side has not been distorted sufficiently to prevent ejection of the illuminant assembly.

6.8 Proving ground test.-Initial production testing shall be conducted in accordance with U.S. Army Test and Evaluation Command Initial Production Test Plan No. CART-40GL-VAR.

6.9 Proving ground test summary:

TEST	SAMPLE SIZE	REQUIREMENTS
Functioning		
(1) First article		
a. Hot	80	See 3.4.5 & 4.2.4
b. Ambient	280	See 3.4.6 & 4.2.4
c. Cold	80	See 3.4.7 & 4.2.4
(2) First 3 lots		
a. Hot	44	See 3.4.5 & 4.3.3.8.4
b. Ambient	176	See 3.4.6 & 4.3.3.8.4
c. Cold	44	See 3.4.7 & 4.3.3.8.4
(3) Regular Production		
a. Hot	26	See 3.4.5 & 4.3.3.11.4
b. Ambient	80	See 3.4.6 & 4.3.3.11.4
c. Cold	26	See 3.4.7 & 4.3.3.11.4

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1310-A007

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